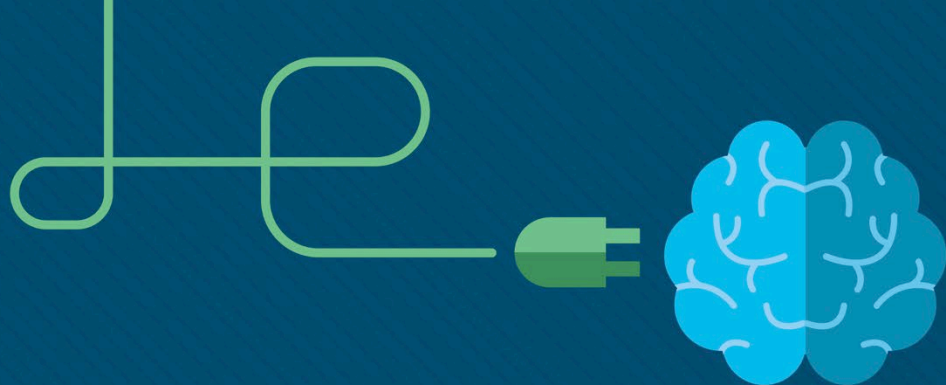




Week 13

RH124 Chapter 12: Install and Update Software Packages



Package Management

1. RPM Package Management

- Red Hat's package management system is based upon the file format used to package the software files used within that distribution, known as the **RPM Package Manager**. Although invented at Red Hat, RPM is used for managing software for many different distributions and is the baseline software management system defined by the **Linux Standards Base**.
- RPM files are available in two formats: .src.rpm files and .rpm files. Unless you are compiling the software yourself or viewing the source code of the software, you don't need to be concerned with the .src.rpm files. These files contain the source code that is used to build the binary .rpm files. The binary .rpm files are the ones that contain the compiled code that is ready to use.
- The files that are used by RPM have a specific naming convention to help identify the package, its version number, its release identifier, and which architecture the software was built for. For example, if a file was named:

```
x3270-x11-3.3.6-10.5.el6.i686.rpm
```

- Then critical information could be discovered from just the file name:

```
package_name-version-release.architecture.rpm
```

1. RPM Package Management (Cont'd)

Package Name

x3270-x11-3.3.6-10.5.el6.i686.rpm

Version Number

x3270-x11-3.3.6-10.5.el6.i686.rpm

Release Identifier

x3270-x11-3.3.6-10.5.el6.i686.rpm

Architecture

x3270-x11-3.3.6-10.5.el6.i686.rpm

This version number is set by the original developer of the software. Release identifiers are chosen by the packager (not the software developers) of the software, which is normally the vendor of the distribution.

Code	Architecture
noarch	Packages that do not contain binary code, such as those that may only contain script files
src	Packages that have not been compiled
i686	Packages compiled to run in 32-bit mode on a Pentium 4 or later processor
x86_64	Packages compiled to run in 64-bit mode on either an AMD 64-bit processor or an Intel EM64T 64-bit compatible processor

2.RPM Queries

- Any user can perform RPM *queries* to display information about packages that are currently installed on the system or about RPM files that have yet to be installed (either on the filesystem or reachable by a URL). The difference between doing a query about an installed package versus an RPM file is two-fold:
 - To query a package that is not installed, use the **-p FILE** option. The **FILE** argument specifies the path to a local .rpm file or URL of an .rpm file on the internet (ftp or http).
 - To query an installed package, use only the package name.
- To perform any RPM query, always use the **-q** option with the rpm command. To query for basic package information, use the **-i** option.

```
[root@localhost ~]# rpm -qip /media/Packages/httpd-2.4.6-88.el7.x86_64.rpm
warning: /media/Packages/httpd-2.4.6-88.el7.x86_64.rpm: Header V3 RSA/SHA256 Signature, key ID fd431d51: NOKEY
Name       : httpd
Version    : 2.4.6
Release    : 88.el7
Architecture: x86_64
Install Date: (not installed)
Group      : System Environment/Daemons
Size       : 3887816
License    : ASL 2.0
Signature  : RSA/SHA256, Fri 22 Jun 2018 04:18:19 AM EDT, Key ID 199e2f91fd431d51
Source RPM : httpd-2.4.6-88.el7.src.rpm
Build Date : Fri 22 Jun 2018 01:20:40 AM EDT
Build Host : x86-041.build.eng.bos.redhat.com
Relocations : (not relocatable)
Packager   : Red Hat, Inc. <http://bugzilla.redhat.com/bugzilla>
Vendor     : Red Hat, Inc.
URL        : http://httpd.apache.org/
Summary    : Apache HTTP Server
Description:
The Apache HTTP Server is a powerful, efficient, and extensible
web server.
```

2. RPM Queries (Cont'd)

- Always remember that RPM queries require the use of the -q option for installed packages and the -qp option for uninstalled package files. There are many different types of queries that can be performed, as described in the following table:

Option	Purpose
-a	List all of the installed packages currently on the system
-c	Display a list of configuration files that belong to the package
-d	List the documentation files that belong to the package
-i	Display the package information
-K	Check the package integrity
-l	List all files in the package
--provides	List the capabilities this package provides
-R	List the capabilities that this package requires
--scripts	List the scripts that are used before and after installation of the package
-s	Display the state of each package file as normal, not installed or replaced

2. RPM Queries (Cont'd)

Verifying Scripts

- You can also use the `--scripts` option to query a package to see the scripts it will execute before installing an RPM file from a third party source. It is potentially dangerous to run RPM packages without first verifying the scripts as it will be running under root privileges and may potentially cause a lot of damage.

```
[root@localhost ~]# rpm -qp --scripts /media/Packages/httpd-2.4.6-88.el7.x86_64.rpm
warning: /media/Packages/httpd-2.4.6-88.el7.x86_64.rpm: Header U3 RSA/SHA256 Signature, key ID fd431d51: NOKEY
preinstall scriptlet (using /bin/sh):
# Add the "apache" group and user
/usr/sbin/groupadd -g 48 -r apache 2> /dev/null || :
/usr/sbin/useradd -c "Apache" -u 48 -g apache \
    -s /sbin/nologin -r -d /usr/share/httpd apache 2> /dev/null || :
postinstall scriptlet (using /bin/sh):

if [ $1 -eq 1 ] ; then
    # Initial installation
    systemctl preset httpd.service htcacheclean.service >>/dev/null 2>&1 || :
fi
preuninstall scriptlet (using /bin/sh):

if [ $1 -eq 0 ] ; then
    # Package removal, not upgrade
    systemctl --no-reload disable httpd.service htcacheclean.service > /dev/null 2>&1 || :
    systemctl stop httpd.service htcacheclean.service > /dev/null 2>&1 || :
fi
postuninstall scriptlet (using /bin/sh):
```

2. RPM Queries (Cont'd)

Verifying the Integrity of a Package

- In addition to checking the scripts ran during the RPM install, you may also wish to verify the integrity of the package file using the -K option as follows:

```
[root@localhost ~]# rpm -K /media/Packages/httpd-2.4.6-88.el7.x86_64.rpm  
/media/Packages/httpd-2.4.6-88.el7.x86_64.rpm: rsa sha1 (md5) pgp md5 OK
```

- Prior to this, you will need to run the following commands to import the keys which are also found in /etc/pki/rpm-gpg/* and verify the list of keys imported:

```
[root@localhost ~]# rpm --import /etc/pki/rpm-gpg/*  
[root@localhost ~]# rpm -qa gpg-pubkey*  
gpg-pubkey-37017186-45761324  
gpg-pubkey-f21541eb-4a5233e7  
gpg-pubkey-db42a60e-37ea5438  
gpg-pubkey-42193e6b-4624eff2  
gpg-pubkey-2fa658e0-45700c69  
gpg-pubkey-897da07a-3c979a7f  
gpg-pubkey-fd431d51-4ae0493b
```

2. RPM Queries (Cont'd)

Querying for Installed Packages

- If a package has been installed, it will return the package name if queried by name as follows:

```
[root@localhost ~]# rpm -q vsftpd
vsftpd-3.0.2-25.el7.x86_64
```

- If a package is not installed, the output will be as follows:

```
[root@localhost ~]# rpm -q httpd
package httpd is not installed
```

- The * character may be used to query if the exact package name is not known, example,

```
[root@localhost ~]# rpm -qa python*
python-pycurl-7.19.0-19.el7.x86_64
python-dateutil-1.5-7.el7.noarch
python-chardet-2.2.1-1.el7_1.noarch
python-ipaddress-1.0.16-2.el7.noarch
python-inotify-0.9.4-4.el7.noarch
python-setuptools-0.9.8-7.el7.noarch
python-urlgrabber-3.10-9.el7.noarch
python-firewall-0.5.3-5.el7.noarch
```

- To get an alphabetical list of all installed packages, execute the **rpm -qa | sort** command.
- To get a chronological list of all packages, execute the **rpm -qa --last** command

3. Installing Packages with RPM

- When using rpm to install software packages, the installation will fail if any dependent software package are not already installed. For example, if you try to install httpd-2.4.6-88.el7.x86_64.rpm, you will receive the following error:

```
[root@localhost ~]# rpm -i /media/Packages/httpd-2.4.6-88.el7.x86_64.rpm
error: Failed dependencies:
    /etc/mime.types is needed by httpd-2.4.6-88.el7.x86_64
    httpd-tools = 2.4.6-88.el7 is needed by httpd-2.4.6-88.el7.x86_64
    libapr-1.so.0()(64bit) is needed by httpd-2.4.6-88.el7.x86_64
    libaprutil-1.so.0()(64bit) is needed by httpd-2.4.6-88.el7.x86_64
```

- This is due to missing software packages that the httpd-2.4.6-88.el7.x86_64.rpm package is dependent upon.
- To resolve the above dependencies, the administrator would have to investigate the software packages that provides the /etc/mime.types and the libraries that provides libapr-1.so.0()(64bit) and libaprutil-1.so.0()(64bit) modules. In the event of circular dependencies, the packages need to be installed together by executing the rpm -i command with multiple software package file names as arguments.

3. Installing Packages with RPM (Cont'd)

- Following on from the previous example of installing httpd package, upon investigation, you will find that the library apr-util-1.5.2-6.el7.x86_64.rpm and apr-1.4.8-3.el7_4.1.x86_64.rpm provides libaprutil-1.so.0()(64bit) and libapr-1.so.0()(64bit) respectively.
- You may now attempt to install the httpd software package by installing these packages together in the rpm -i command. The last package mailcap being the software package that provides /etc/mime.types

```
[root@localhost Packages]# rpm -i httpd-2.4.6-88.el7.x86_64.rpm httpd-tools-2.4.6-88.el7.x86_64.rpm  
apr-1.4.8-3.el7_4.1.x86_64.rpm apr-util-1.5.2-6.el7.x86_64.rpm mailcap-2.1.41-2.el7.noarch.rpm
```

4. Erasing Packages with RPM

- The rpm command can be used to erase (remove) packages from the system with the -e option. However, similar to package installation, dependency issues may arise. If there are software packages that depends on the software package being removed, then these dependent software packages need to be removed first.

- For example, removing the httpd-tools package will result in the following error:

```
[root@localhost Packages]# rpm -e httpd-tools
error: Failed dependencies:
    httpd-tools = 2.4.6-88.el7 is needed by (installed) httpd-2.4.6-88.el7.x86_64
```

- To remove the httpd package, you can remove both the httpd and httpd-tools in the same command as follows:

```
[root@localhost Packages]# rpm -e httpd httpd-tools apr apr-util mailcap
[root@localhost Packages]#
```

5. Updating Packages with RPM

- To update a package with a new version or newer release, use the rpm command with either the -U or the -F option. Dependency issues may still arise when updating packages. Installing an updated package often requires the installation of updates to the packages that it depends upon.
- The -U option can be used with the rpm command to either install or update a package.
- However, the -F option can only be used if the package is currently installed.

6. Using rpm2cpio

- The rpm2cpio command accepts an .rpm file as an argument (or reads data that is in a package format from standard input) and outputs a cpio archive. This capability can be used to list the contents of an .rpm file or to extract one or more files from within the original .rpm file.
- For example, to list the contents of the telnet-server-0.17-64.el7.x86_64.rpm file, execute the following command. The output shows the files contained within the .rpm file and the total number of blocks that those files would use if extracted:

```
[root@localhost Packages]# rpm2cpio telnet-server-0.17-64.el7.x86_64.rpm | cpio -t
./usr/lib/systemd/system/telnet.socket
./usr/lib/systemd/system/telnet@.service
./usr/sbin/in.telnetd
./usr/share/man/man5/issue.net.5.gz
./usr/share/man/man8/in.telnetd.8.gz
./usr/share/man/man8/telnetd.8.gz
113 blocks
```

- Using the cpio command to extract only selected files from the package file is useful in situations where only a key package file is accidentally removed and reinstalling the entire package is overkill.

6. Using rpm2cpio (Cont'd)

- To copy these files from the .rpm file, use the -imud option to the cpio command as follows:

```
[root@localhost ~]# rpm2cpio /media/Packages/telnet-server-0.17-64.el7.x86_64.rpm | cpio -imud  
113 blocks
```

```
[root@localhost ~]# ls -R usr
```

```
usr:  
lib sbin share
```

```
usr/lib:  
systemd
```

```
usr/lib/systemd:  
system
```

```
usr/lib/systemd/system:  
telnet@.service telnet.socket
```

```
usr/sbin:  
in.telnetd
```

```
usr/share:  
man
```

```
usr/share/man:  
man5 man8
```

```
usr/share/man/man5:  
issue.net.5.gz
```

```
usr/share/man/man8:  
in.telnetd.8.gz telnetd.8.gz
```

Option	Purpose
-i	Extract
-m	Retain the original modification times on the files
-u	Unconditionally replace any existing files
-d	Create any parent directories of the files contained within the archive

7. Managing Packages with YUM

Please note that DNF (Dandified YUM) has replaced YUM on Red Hat. The YUM commands listed in this section are actually shortcut links to the corresponding DNF commands. Syntax between YUM and DNF commands are similar.

- While an administrator can manage software directly with the rpm command, other higher level commands are available that make software management significantly easier. The yum command can locate and download packages on the Internet and resolve dependencies automatically.
- Typically, the yum command is configured by editing the /etc/yum.conf file and the files found in the /etc/yum.repos.d directory. These configuration files are used to specify servers (the repositories) on the Internet where the yum command can obtain the RPM files automatically. If the system does not have networking enabled or doesn't have access to the Internet, then it is still possible to configure yum to install software packages from a media such as CD or DVD.
- Yum can also be used to determine which package owns a file because the yum command searches the database of the repository server(s).

```
[root@localhost bin]# yum provides /usr/bin/bzip2
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: mirror.0x.sg
* centosplus: mirror.0x.sg
* extras: mirror.0x.sg
* updates: mirror.0x.sg
bzip2-1.0.6-13.el7.x86_64 : A file compression utility
Repo                : base
Matched from:
Filename            : /usr/bin/bzip2
```

7. Managing Packages with YUM (Cont'd)

- The other distinct advantage of using the yum command over the rpm command is the ability to search for a package in the repositories, based upon the name or description of the package.

```
[root@localhost ~]# yum search terminal
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
 * base: centos.usonyx.net
 * centosplus: centos.usonyx.net
 * extras: centos.usonyx.net
 * updates: centos.usonyx.net
===== N/S matched: terminal =====
emacs-terminal.noarch : A desktop menu item for GNU Emacs terminal.
gedit-plugin-terminal.x86_64 : gedit terminal plugin
gnome-terminal.x86_64 : Terminal emulator for GNOME
gnome-terminal-nautilus.x86_64 : GNOME Terminal extension for Nautilus
kbd.x86_64 : Tools for configuring the console (keyboard, virtual terminals, etc.)
konsole.x86_64 : KDE Terminal emulator
minicom.x86_64 : A text-based modem control and terminal emulation program
ncurses-base.noarch : Descriptions of common terminals
ncurses-term.noarch : Terminal descriptions
perl-TermReadKey.x86_64 : A perl module for simple terminal control
perl-Text-CharWidth.x86_64 : Get number of occupied columns of a string on terminal
screen.x86_64 : A screen manager that supports multiple logins on one terminal
tmux.x86_64 : A terminal multiplexer
```

8. Installing Packages with YUM

- The yum utility is the preferred command to install packages since it will resolve dependency issues automatically. For example, to install the httpd package, an administrator can execute the following:

```
[root@localhost ~]# yum install httpd
```

```
Resolving Dependencies
--> Running transaction check
---> Package httpd.x86_64 0:2.4.6-90.el7.centos will be installed
--> Processing Dependency: httpd-tools = 2.4.6-90.el7.centos for package: httpd-2.4.6-90.el7.centos.x86_64
Dependencies Resolved
--> Processing Dependency: /etc/mime.types for package: httpd
--> Processing Dependency: libaprutil-1.so.0()(64bit) for package: httpd
--> Processing Dependency: libapr-1.so.0()(64bit) for package: httpd
--> Running transaction check
---> Package apr.x86_64 0:1.4.8-5.el7 will be installed
---> Package apr-util.x86_64 0:1.5.2-6.el7 will be installed
---> Package httpd-tools.x86_64 0:2.4.6-90.el7.centos will be installed
---> Package mailcap.noarch 0:2.1.41-2.el7 will be installed
--> Finished Dependency Resolution

Installing:
Package Arch Version Repository Size
-----
httpd x86_64 2.4.6-90.el7.centos base 2.7 M

Installing for dependencies:
Package Arch Version Repository Size
-----
apr x86_64 1.4.8-5.el7 base 103 k
apr-util x86_64 1.5.2-6.el7 base 92 k
httpd-tools x86_64 2.4.6-90.el7.centos base 91 k
mailcap noarch 2.1.41-2.el7 base 31 k

Transaction Summary
-----
Install 1 Package (+4 Dependent packages)

Total download size: 3.0 M
Installed size: 10 M
Is this ok [y/d/N]: _
```

8. Installing Packages with YUM (Cont'd)

- At the “Is this ok?” prompt, you can either respond “y” for yum to proceed with the installation of the dependent software packages and httpd, or you can respond with “d” to just download the packages without installation or “N” to cancel the installation.
- To skip the “Is this ok?” prompt, you can enter the yum install command with the -y option. (top right)
- You can also install multiple software packages together by listing the packages as arguments of the yum install command. (middle right)
- Finally, you can all install predefined groups of software packages using yum groups. Use the yum grouplist command to list all of the groups that are installed on the system, as well as groups that are available. (bottom right)

```
[root@localhost ~]# yum install -y httpd
```

```
[root@localhost ~]# yum install telnet vsftpd sshd
```

```
[root@localhost ~]# yum grouplist | head -n 15
Loaded plugins: fastestmirror, langpacks
There is no installed groups file.
Maybe run: yum groups mark convert (see man yum)
Loading mirror speeds from cached hostfile
* base: centos.usonyx.net
* centosplus: centos.usonyx.net
* extras: centos.usonyx.net
* updates: centos.usonyx.net
Available Environment Groups:
  Minimal Install
  Compute Node
  Infrastructure Server
  File and Print Server
  Basic Web Server
  Virtualization Host
  Server with GUI
  GNOME Desktop
```

8. Installing Packages with YUM (Cont'd)

- To see details about a specific group, use the `yum groupinfo` command.
- Use the `yum groupinstall` command to install a yum group (example below)

```
[root@localhost ~]# yum groupinstall "Security Tools"
Loaded plugins: fastestmirror, langpacks
There is no installed groups file.
Maybe run: yum groups mark convert (see man yum)
Loading mirror speeds from cached hostfile
 * base: centos.usonyx.net
 * centosplus: centos.usonyx.net
 * extras: centos.usonyx.net
 * updates: centos.usonyx.net
Resolving Dependencies
--> Running transaction check
--> Package scap-security-guide.noarch 0:0.1.43-13.el7.centos will be installed
--> Processing Dependency: openscap-scanner >= 1.2.5 for package: scap-security-guide-0.1.43-13.el7.centos.noarch
--> Processing Dependency: xml-common for package: scap-security-guide-0.1.43-13.el7.centos.noarch
--> Running transaction check
--> Package openscap-scanner.x86_64 0:1.2.17-4.el7 will be installed
--> Processing Dependency: openscap(x86-64) = 1.2.17-4.el7 for package: openscap-scanner-1.2.17-4.el7.x86_64
--> Processing Dependency: libxslt.so.1()(64bit) for package: openscap-scanner-1.2.17-4.el7.x86_64
--> Processing Dependency: libopenscap.so.8()(64bit) for package: openscap-scanner-1.2.17-4.el7.x86_64
```

```
[root@localhost ~]# yum groupinfo "Minimal Install"
Loaded plugins: fastestmirror, langpacks
There is no installed groups file.
Maybe run: yum groups mark convert (see man yum)
Loading mirror speeds from cached hostfile
 * base: centos.usonyx.net
 * centosplus: centos.usonyx.net
 * extras: centos.usonyx.net
 * updates: centos.usonyx.net

Environment Group: Minimal Install
Environment-Id: minimal
Description: Basic functionality.
Mandatory Groups:
  +core
Optional Groups:
  +debugging
```

9. Removing Packages with YUM

- Removing a software package with yum will also have similar benefits of being able to automatically resolve any dependency issues. When removing a package that has other packages that depend upon it, yum will offer to remove the dependencies.

```
[root@localhost ~]# yum remove httpd-tools
Loaded plugins: fastestmirror, langpacks
Resolving Dependencies
--> Running transaction check
--> Package httpd-tools.x86_64 0:2.4.6-90.el7.centos will be erased
--> Processing Dependency: httpd-tools = 2.4.6-90.el7.centos for package: httpd-2.4.6-90.el7.centos.x86_64
--> Running transaction check
--> Package httpd.x86_64 0:2.4.6-90.el7.centos will be erased
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package                Arch             Version           Repository        Size
=====
Removing:
httpd-tools            x86_64           2.4.6-90.el7.centos @base             169 k
Removing for dependencies:
httpd                  x86_64           2.4.6-90.el7.centos @base             9.4 M

Transaction Summary
=====
Remove 1 Package (+1 Dependent package)

Installed size: 9.5 M
Is this ok [y/N]:
```

- You can also remove an entire yum group using the yum groupremove command followed by the group name.

10. Determining if a Package has been installed

- To determine if a software package has been installed, use the `yum list <package name>` command. In the example below, `httpd` is determined to be “Installed Packages”.

```
[root@localhost ~]# yum list httpd
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
 * base: centos.shinjiru.com
 * centosplus: centos.shinjiru.com
 * extras: centos.shinjiru.com
 * updates: centos.ipserverone.com
Installed Packages
httpd.x86_64                2.4.6-90.el7.centos      @base
```

- Similarly, the `yum grouplist <group name>` command determines if a yum group is currently installed.
- If a package is not installed, the `yum list` command will display “Available Packages”.

11. Updating Packages with YUM

- Using the `yum list updates` command will cause the system to connect to repositories on the Internet to identify newer versions or releases of packages and list them. By using the `yum update` command, the system will download and update all possible packages after requesting confirmation.
- To update one or more individual packages, the following syntax can be used.

```
yum update [PACKAGE]...
```

- The `yum update` command will also install a new copy of an updated kernel and keep the previous version.

Output of the `yum list updates` command on the right.

```
Updated Packages
GeoIP.x86_64                    1.5.0-14.el7                base
NetworkManager.x86_64          1:1.18.0-5.el7_7.1          updates
NetworkManager-libnm.x86_64    1:1.18.0-5.el7_7.1          updates
NetworkManager-team.x86_64     1:1.18.0-5.el7_7.1          updates
NetworkManager-tui.x86_64      1:1.18.0-5.el7_7.1          updates
abrt.x86_64                     2.1.11-55.el7.centos        base
abrt-addon-ccpp.x86_64         2.1.11-55.el7.centos        base
abrt-addon-kerneloops.x86_64   2.1.11-55.el7.centos        base
abrt-addon-pstoreoops.x86_64   2.1.11-55.el7.centos        base
abrt-addon-python.x86_64       2.1.11-55.el7.centos        base
abrt-addon-vmcore.x86_64       2.1.11-55.el7.centos        base
abrt-addon-xorg.x86_64         2.1.11-55.el7.centos        base
```

12. Configuring repo files to use local / remote iso repositories

- To configure yum to use local / remote iso repositories, do the following:

1. After booting up the CentOS VM and logging in as the root user, insert the centos 9 dvd into the optical drive.

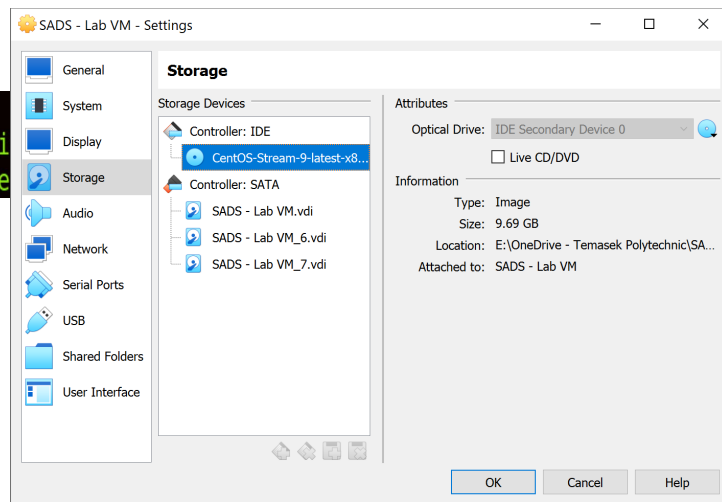
2. Type `mount | grep sr0`

```
[root@sads-lab ~]# mount | grep sr0
/dev/sr0 on /run/media/root/CentOS-Stream-9-BaseOS-x86_64 type iso9660 (ro,nosuid,
ime,nojoliet,check=s,map=n,blocksize=2048,uid=0,gid=0,dmode=500,fmode=400,uhelp
```

3. Create a new repo file name `cdrom.repo` in `/etc/yum.repos.d` directory. Enter the following:

```
[media-appstream]
name=media-appstream
baseurl=file:///run/media/root/CentOS-Stream-9-BaseOS-x86_64/AppStream
gpgcheck=0
enabled=1

[media-baseos]
name=media-baseos
baseurl=file:///run/media/root/CentOS-Stream-9-BaseOS-x86_64/BaseOS
gpgcheck=0
enabled=1
```



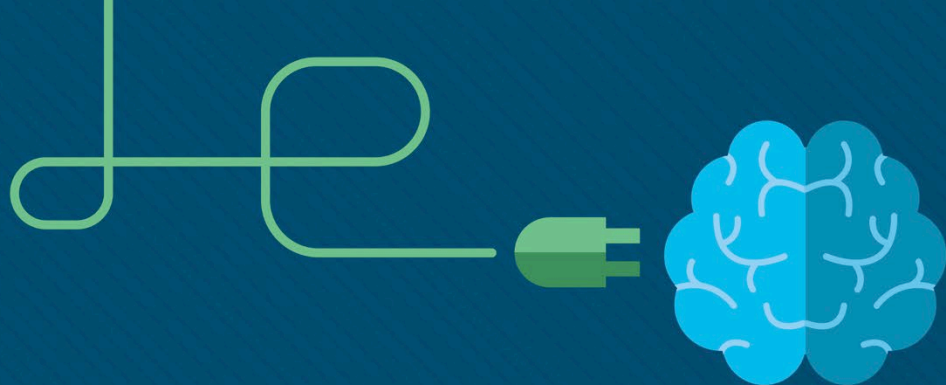
12. Configuring repo files to use local / remote iso repositories (Cont'd)

4. Ensure all other repositories (except the cdrom.repo) are disabled by issuing the following

```
[root@sads-lab ~]# ls /etc/yum.repos.d | grep -v cdrom.repo | xargs -I '{}' sed -i s/enabled\
=1/enabled=0/g '/etc/yum.repos.d/{'
```

5. Execute yum repolist (or dnf repolist to confirm the list of enabled repositories)

```
[root@sads-lab ~]# yum repolist
repo id                repo name
media-appstream         media-appstream
media-baseos             media-baseos
```



Lab

RH124 Chapter 12.9: Lab Install and Update Software Packages



Practice Exercises

1 RPM Package Management

- Identify the RPM package format and explain its components.
- Use `rpm -q` to check if `nano` is installed.
- Query detailed package information for `bash` using `rpm -qi`.
- Verify the integrity of an installed package with `rpm -V`.

2 Installing & Removing RPM Packages

- Install an RPM package from a local file.
- Remove an installed package using `rpm -e`. What happens if dependencies exist?
- Use `rpm -U` to upgrade an existing package.

3 Managing Packages with YUM/DNF

- Use `yum/dnf list installed` to display installed packages.
- Search for a package using `yum/dnf search`.
- Install `htop` using `yum/dnf install`.
- Remove `htop` with `yum/dnf remove`.

4 Configuring YUM/DNF Repositories

- List enabled repositories using `yum/dnf repolist`.
- Add a new repository by creating a `.repo` file in `/etc/yum.repos.d/`.
- Disable all repositories except `cdrom.repo` and verify the changes.